Practice Exam

Philosophy 305: Introduction to Formal Methods

2026-04-15

Question 1: Probability and Conditional Probability

A payment processing company uses an automated system to flag potentially fraudulent transactions. From historical data, 2% of all transactions are fraudulent. The system correctly flags 90% of fraudulent transactions. However, it also incorrectly flags 5% of legitimate transactions.

(a): What is the probability that a randomly selected transaction gets flagged by the system?

(b): A transaction has been flagged. What is the probability that it is actually fraudulent?

(c): Is the event "the transaction is flagged" independent of the event "the transaction is fraudulent"? Justify your answer using the definition of independence.

Question 2: Probability Trees and Independence

A student takes a two-part qualifying exam. The probability of passing Part 1 is 0.7. If they pass Part 1, the probability of passing Part 2 is 0.8 (a confidence boost). If they fail Part 1, the probability of passing Part 2 drops to 0.3.

(a): What is the probability that the student passes both parts?

(b): What is the probability that the student passes exactly one of the two parts?

Question 3: Expected Utility

A small business owner must choose between two expansion plans before learning the economic conditions for the coming year. There are three possible scenarios:

	Downturn (Pr = 0.3)	Stable (Pr = 0.4)	Growth (Pr = 0.3)
Plan X	-90	80	100
Plan Y	10	30	50

Profits are in thousands of dollars.

(a): Which plan has higher expected monetary value? Show the calculation for each.

(b): Does either plan dominate the other (strictly or weakly)? Explain briefly.

(c): The owner currently has \$100,000 in savings and has a utility function $U(\text{total wealth}) = \sqrt{\text{total wealth in thousands}}$. So, for example, if the owner chooses Plan X and a Downturn occurs, total wealth is \$10,000 and utility is $\sqrt{10}$. Which plan should the owner choose according to expected utility? Does this differ from your answer in part (a)?

Question 4: Strategic-Form Games

Consider the following two-player game, where payoffs are listed as (Row, Column):

	A	B	C	D
T	(4, 2)	(2, 1)	(3, 3)	(1, 0)
M	(2, 3)	(4, 0)	(1, 2)	(3, 1)
B	(1, 1)	(3, 2)	(2, 4)	(0, 0)

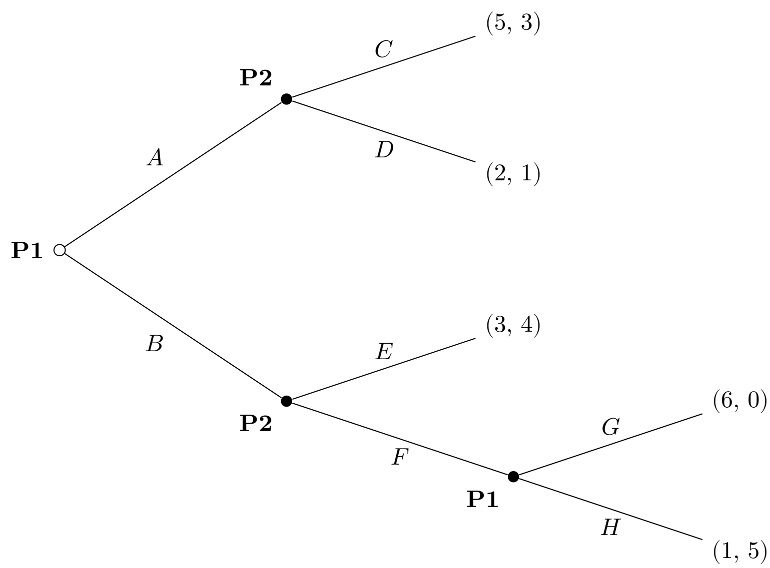
(a): In the original game, does any strategy strictly dominate another for either player? State all dominance relations you find.

(b): Use iterated elimination of strictly dominated strategies to simplify the game. Show each round of elimination clearly, stating which strategy is eliminated and why.

(c): Find all pure-strategy Nash equilibria of the original game. Use the best-response method and show your work.

Question 5: Dynamic Games and Backward Induction

Consider the following game in extensive form. Player 1 moves first, choosing **A** or **B**. Player 2 observes Player 1's choice and responds. If Player 1 chose A, Player 2 chooses **C** or **D**. If Player 1 chose B, Player 2 chooses **E** or **F**. If Player 2 chooses F, then Player 1 gets a second move, choosing **G** or **H**. Payoffs are (Player 1, Player 2):



(a): How many strategies does each player have? List them all. (Remember: a strategy specifies a choice at every decision node where that player moves, including nodes that may not be reached.)

(b): Solve this game using backward induction. State clearly what happens at each step and what the final outcome is.

(c): Player 2 would prefer the outcome (3, 4) to the backward-induction outcome. Suppose Player 2 threatens to play D (rather than C) after A, hoping to push Player 1 toward B. Is this threat credible? Explain.

(d): Write out the strategic (normal) form of this game. That is, construct the payoff table with Player 1's strategies as rows and Player 2's strategies as columns.

Question 6: Mixed Strategies

Consider the following game:

	Left	Right
Up	(3, 1)	(1, 2)
Down	(1, 3)	(2, 0)

This game has no pure-strategy Nash equilibrium. Find the mixed-strategy Nash equilibrium. That is, find the probability with which Row plays Up and the probability with which Column plays Left such that neither player can improve their expected payoff. Also calculate each player's expected payoff in the equilibrium.